

Questions for the Student Report

1. Give the full scientific name, family, NCBI accession/version, database source, and complete plastid-genome size of your selected organism.

My selected organism is *Dracaena draco*, commonly known as the dragon tree. It belongs to the family Asparagaceae. The plastid genome accession is NC_048492.1, and the database source is NCBI RefSeq. The complete plastid genome is 155,422 bp long. The genome size was also obtained from my Galaxy FASTA Statistics result, where the scaffold number of base pairs was 155,422. NCBI identifies NC_048492.1 as the complete *D. draco* chloroplast genome.

2. What evidence shows that the sequence is a complete plastid/chloroplast genome rather than a barcode marker, genome fragment, or nuclear sequence?

The NCBI record specifically identifies NC_048492.1 as “*Dracaena draco* chloroplast, complete genome.” The sequence is also 155,422 bp, which is much larger than a typical short DNA barcode marker. It contains the expected complete chloroplast organization, including the LSC, SSC, and two IR regions, as well as genes involved in photosynthesis, transcription, translation, and other plastid functions. These features support that it is a complete chloroplast genome rather than a small barcode or nuclear DNA fragment.

3. Describe the overall organization of the plastid genome. Does it contain the common LSC-IR-SSC-IR arrangement? Give the sizes of these regions when available.

Yes. The *D. draco* plastid genome has the common LSC–IR–SSC–IR arrangement. The LSC is 83,942 bp, the SSC is 18,472 bp, and each of the two IR regions is 26,504 bp. Together, these regions make up the complete 155,422-bp genome.

4. Summarize the annotated gene content: total genes, protein-coding genes, tRNA genes, rRNA genes, and pseudogenes. Explain why genes located in the inverted-repeat regions may appear in two copies.

The genome contains 132 annotated genes, consisting of 86 protein-coding genes, 38 tRNA genes, and 8 rRNA genes. The study counted 113 unique genes because some genes occur twice in the two IR regions. No major pseudogenes were reported in the published characterization of this plastid genome. Genes in the IR regions can appear in two copies because the two IR regions are repeated sections of the chloroplast genome. Therefore, a gene located within one IR is usually also present in the corresponding position of the other IR.

5. Choose at least eight protein-coding plastid genes from different functional groups. List each gene and briefly explain its biological function.

Gene	Functional group	Biological function
<i>psaA</i>	Photosystem I	Encodes a major core protein of Photosystem I involved in light-driven electron transfer.
<i>psbA</i>	Photosystem II	Encodes the D1 reaction-center protein of Photosystem II, which is important in photosynthetic electron transport.
<i>atpA</i>	ATP synthase	Encodes a subunit of ATP synthase involved in producing ATP during photosynthesis.
<i>petA</i>	Cytochrome <i>b6f</i>	Encodes cytochrome <i>f</i> , which participates in electron transport between Photosystem II and Photosystem I.
<i>rbcL</i>	Carbon fixation	Encodes the large subunit of Rubisco, the enzyme involved in fixing carbon dioxide during the Calvin cycle.
<i>rpoB</i>	RNA polymerase	Encodes a subunit of the plastid-encoded RNA polymerase used for transcription of plastid genes.
<i>matK</i>	RNA processing	Encodes a maturase involved in the processing/splicing of certain group II introns.
<i>ndhA</i>	NADH dehydrogenase	Encodes a subunit of the chloroplast NADH dehydrogenase-like complex involved in electron transport.

6. Identify important RNA and RNA-processing features. Include the rRNA genes, examples of tRNA genes, and at least two genes with introns if present in your genome.

The rRNA genes include *rrn4.5*, *rrn5*, *rrn16*, and *rrn23*, with copies associated with the two IR regions. For example, NCBI annotates the 5S rRNA and 4.5S rRNA genes on NC_048492.1. Examples of tRNA genes include *trnA*, *trnC*, *trnD*, *trnE*, *trnF*, *trnG*, *trnH*, *trnK*, *trnL*, *trnM*, and *trnS*. These genes produce tRNAs needed to translate plastid-encoded proteins. Several genes contain introns. Examples include *rpoC1* and *rpl16*, while *clpP* and *ycf3* contain two introns. Introns need to be removed from the RNA transcript so that the mature RNA can function correctly.

7. Describe any pseudogenes, gene losses, duplications, rearrangements, or other unusual features reported for your plastid genome. If none are reported, state this clearly.

The published characterization did not report major pseudogenes or unusual large-scale rearrangements in the *D. draco* plastid genome. The main duplication is associated with the two IR regions, where genes such as the rRNA genes occur in two copies. The study also identified 77 simple sequence repeats (SSRs).

8. What is the GC content of your plastid genome? Based on your Galaxy results and annotation, describe two other notable sequence or structural observations.

The GC content of the *D. draco* plastid genome is 37.60%, which is the same value shown in my Galaxy FASTA Statistics result. The published study also reports a total GC content of 37.6%.

Two other notable observations are:

1. The genome has a typical quadripartite structure, consisting of one LSC, one SSC, and two IR regions.
2. The genome contains 132 annotated genes and 77 SSRs, showing that the plastid genome contains both conserved functional genes and repetitive sequence features.

9. Compare plastid and mitochondrial genomes.

Feature	Plastid genome	Mitochondrial genome
Location	Inside chloroplasts/plastids	Inside mitochondria
Main role	Photosynthesis and other plastid functions	Cellular respiration and energy production

Important genes	<i>psa, psb, rbcL, atp, pet, rpo, etc.</i>	Mainly respiratory-chain, ATP-production, and mitochondrial gene-expression genes
Typical organization in land plants	Often has LSC, SSC, and two IRs	Much more variable in structure
Genome size	Usually relatively conserved in land plants	Often much more variable, especially in plants
Gene content	Generally more gene-rich than mitochondrial genomes	Usually has fewer genes but can have complicated genome structures
Photosynthesis	Contains genes directly involved in photosynthesis	Does not carry the main photosynthetic genes
Structural variation	Usually relatively conserved in land plants	Plant mitochondrial genomes can have extensive rearrangements
Inheritance	Usually maternal in angiosperms, but exceptions occur	Usually maternal in angiosperms, but exceptions also occur
Evolutionary behavior	Generally relatively conserved and useful for plant phylogeny	Often more structurally dynamic, especially in plants

10. Plastid vs. Mitochondrial Genome Comparison

Feature	Plastid genome	Mitochondrial genome
Cellular location	Located inside plastids, mainly chloroplasts	Located inside mitochondria
Main biological functions	Photosynthesis, carbon fixation, and plastid gene expression	Cellular respiration, ATP production, and mitochondrial gene expression
Typical genome organization	Usually a circular, quadripartite genome with LSC–IR–SSC–IR regions	Usually more structurally variable; plant mitochondrial genomes can occur as complex, recombining DNA molecules

Relative genome size	Usually small and relatively conserved; <i>D. draco</i> plastid genome is 155,422 bp	Usually much more variable in plants and can be considerably larger than the plastid genome
Gene content	Contains genes for photosynthesis, transcription, translation, and other plastid functions, such as <i>psa</i> , <i>psb</i> , <i>rbcL</i> , and <i>rpo</i>	Contains genes mainly associated with respiration, oxidative phosphorylation, and mitochondrial gene expression
Copy number	Multiple copies may occur within a plastid and cell	Multiple copies may occur within mitochondria and cells
Inheritance	Usually maternally inherited in flowering plants, but exceptions occur	Usually maternally inherited in flowering plants, but exceptions occur
Recombination / structural change	Generally conserved with relatively few large structural changes	More frequent recombination and extensive structural variation are common in plant mitochondrial genomes
Mutation / substitution pattern	Generally relatively conserved, with slower overall sequence change in many plastid regions	Coding regions can be relatively conserved, but mitochondrial genomes often show greater structural and non-coding variation
Common research applications	Species identification, DNA barcoding, phylogenetics, plant evolution, population studies, and conservation genetics	Plant evolution, phylogenetics, population genetics, mitochondrial inheritance, and studies of cytoplasmic genome evolution